



Network Design Proposal

CougarTech Living Hall—Multi-Campus Expansion

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Introduction

For this project (titled CougarTech Living Hall), a comprehensive network design must be developed to serve all future students at the University of Houston. This facility is part of a strategic initiative to connect student living across three campuses: UH Main, UH Sugar Land, and UH Clear Lake. CougarTech Living Hall is designed as an **8-story building** that will provide residential housing, administrative space, and multipurpose areas for student use.

Floor 1 of the building houses the main lobby, administrative offices, and the building's primary network infrastructure room. **Floor 2** features conference rooms, study areas, and multipurpose spaces designed to support student collaboration and learning. **Floors 3 to 8** consist of residential units, specifically 27 two-bedroom apartments on each level. These upper floors will be the primary focus of the network distribution, as they require high-density wireless and wired network coverage to support the large number of student occupants.

This proposal includes a **full-scale design** of both wired and wireless networks for the residence hall, with coverage maps for each floor and dedicated VLANs for students, staff, and guests and a network backbone that covers the entire area. Wi-Fi access will be provided using both 2.4 GHz and 5 GHz frequency bands to ensure compatibility and speed across a wide range of devices. All network infrastructure components will be sourced exclusively from Cisco, in compliance with project specifications.

The design will begin with the main distribution facility on Floor 1, which will serve as the core of the network. From there, the network will extend through intermediate distribution points to provide full wired and wireless coverage across the building. This includes access points, switches, servers, and firewalls to ensure security, scalability, and consistent performance for all building residents.



Recommendation

We are designing the networks in **seven** network architecture components: local area networks (LANs), building backbone networks, campus backbones that connect buildings, wide area network (WANs) that connect campuses, Internet access, e-commerce edge, and data centers.

Creating a network infrastructure for a student living complex spanning three campuses presents a unique set of challenges and opportunities. You need a robust, scalable, and manageable solution that can handle the high density of users, diverse application needs (streaming, gaming, research, etc.), and potential future growth.

Out of the two options between 3-Tier Network Architecture and Spine/Leaf Network Architecture, the 3-Tier network was selected due to well-established and widely understood architecture, cost effective for smaller, less demanding networks and the 3 layered approach can offer granular control over network policies at different layers offering better security.

For the UH CougarTech Living Hall planned in the UH Main, Clear Lake, and Sugar Land campuses, we will be using identical network design for each of the campuses and connect them using a WAN leased from a common carrier like AT&T. The WAN is made for private use and carries traffic between the different campuses. The total budget to execute the network plan is \$500,000. Pricing for the materials will be sourced from cdw.com.

The 3-Tier architecture consists of three layers:

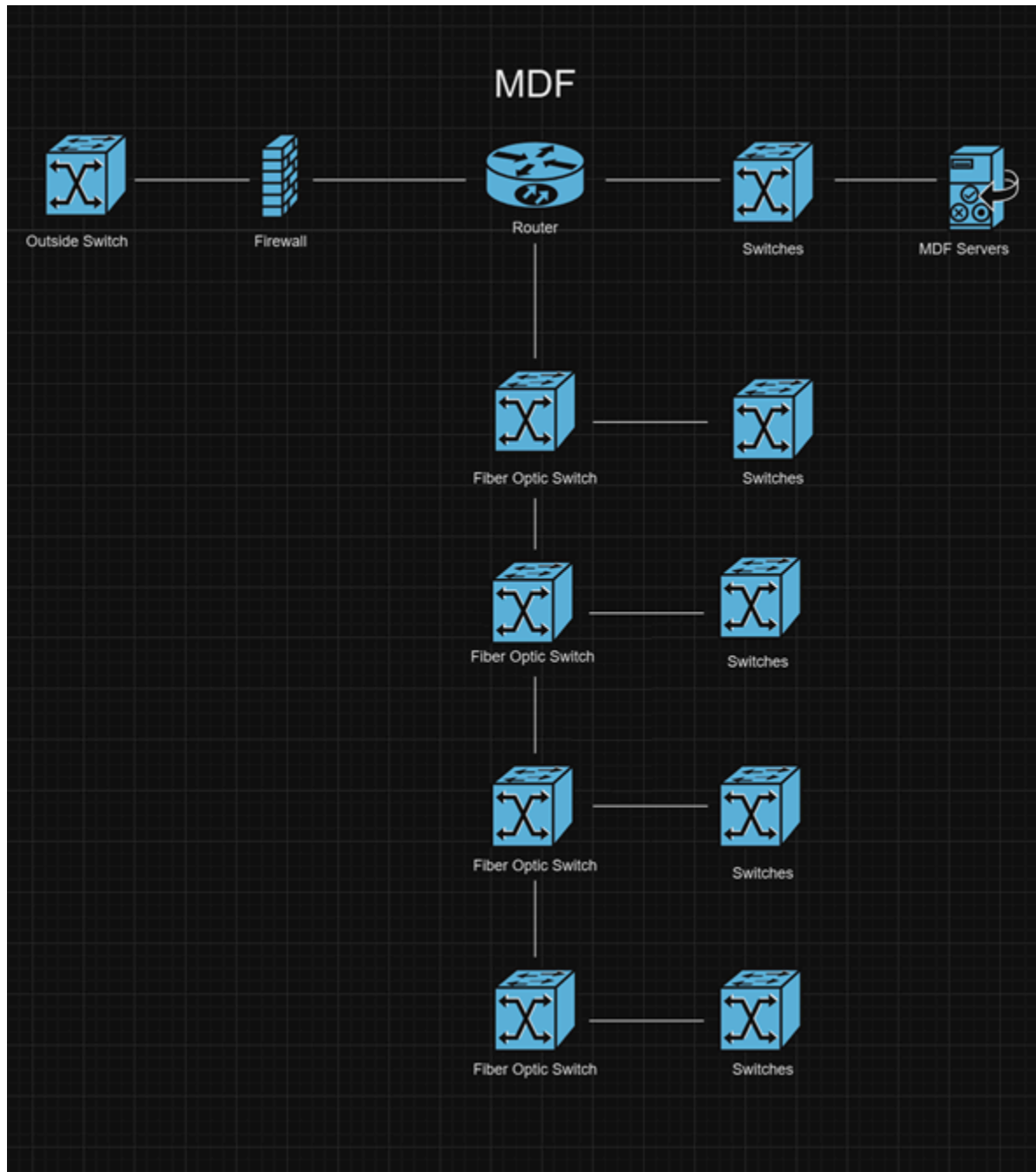
- **Access Layer:** Where end-user devices (laptops, smartphones, gaming consoles) connect to the network; usually involves switches providing connectivity/basic security features.
- **Aggregation (or Distribution) Layer:** This layer aggregates traffic from the access layer switches and provides policy enforcement, routing between VLANs, and redundancy.
- **Core Layer:** This is the high-speed backbone of the network, responsible for efficient and reliable transport of traffic between different parts of the network, including potentially between campuses.



Diagrams / Floor Plans

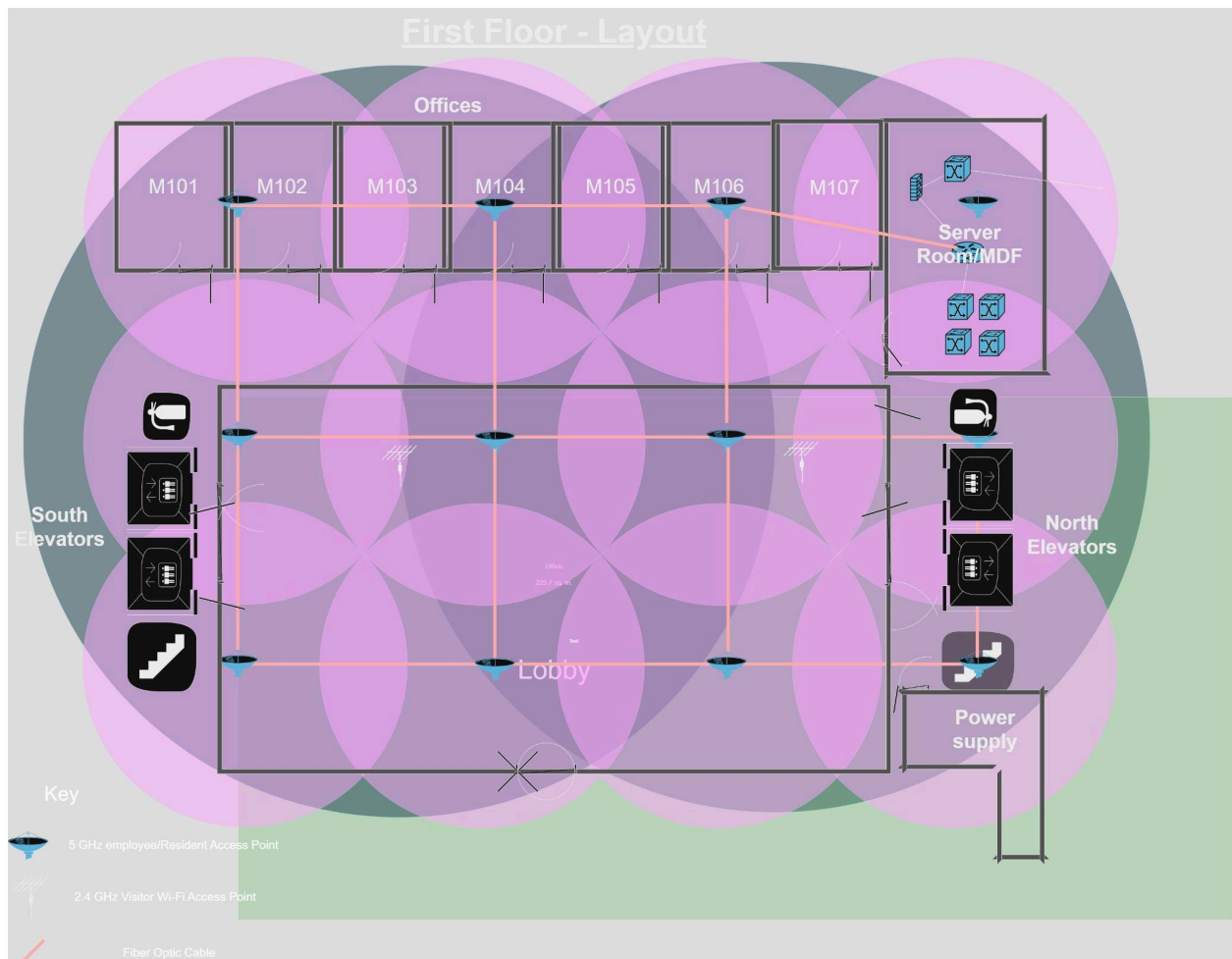
Network Diagrams

Data Center



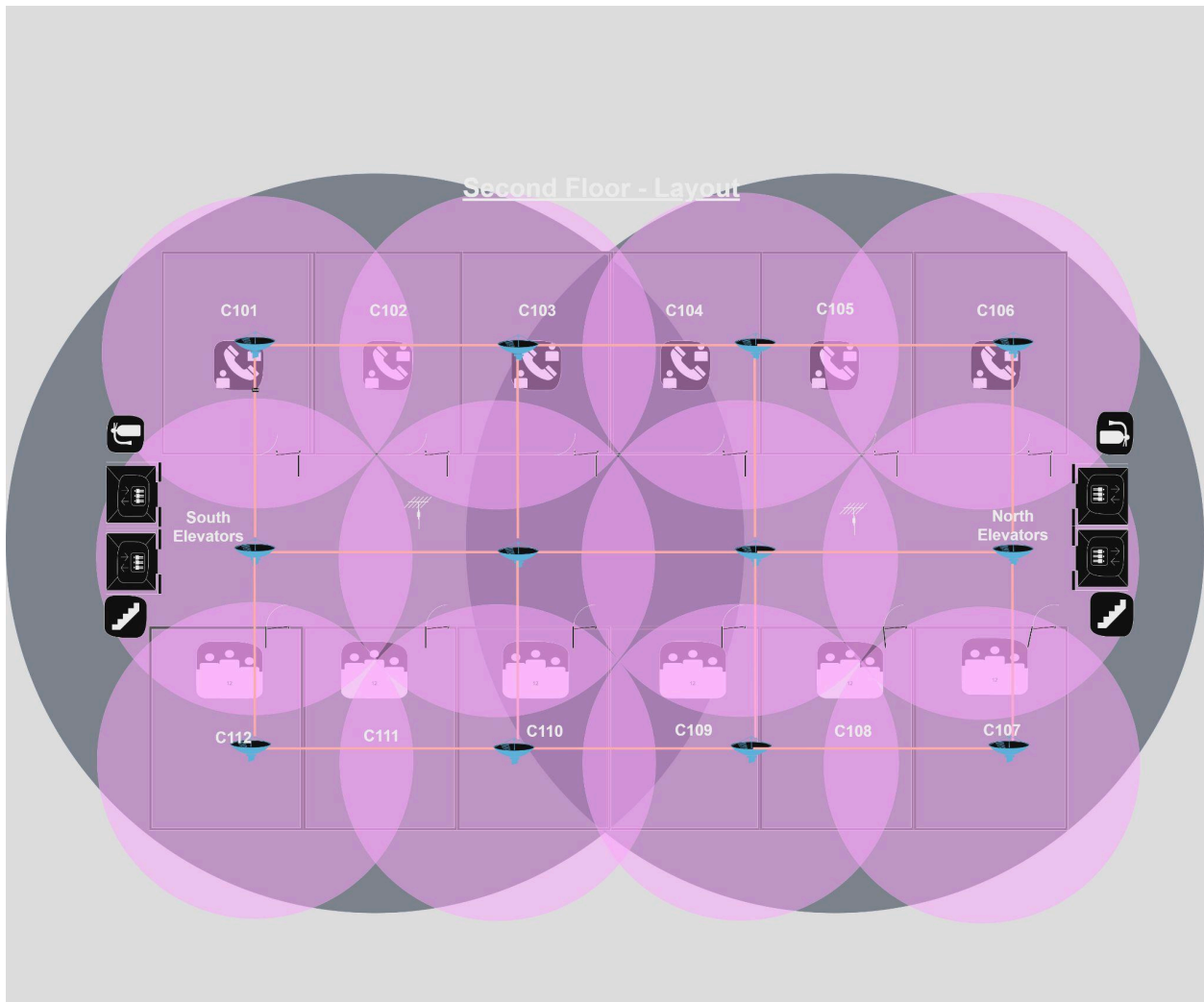


Layout of Floor 1



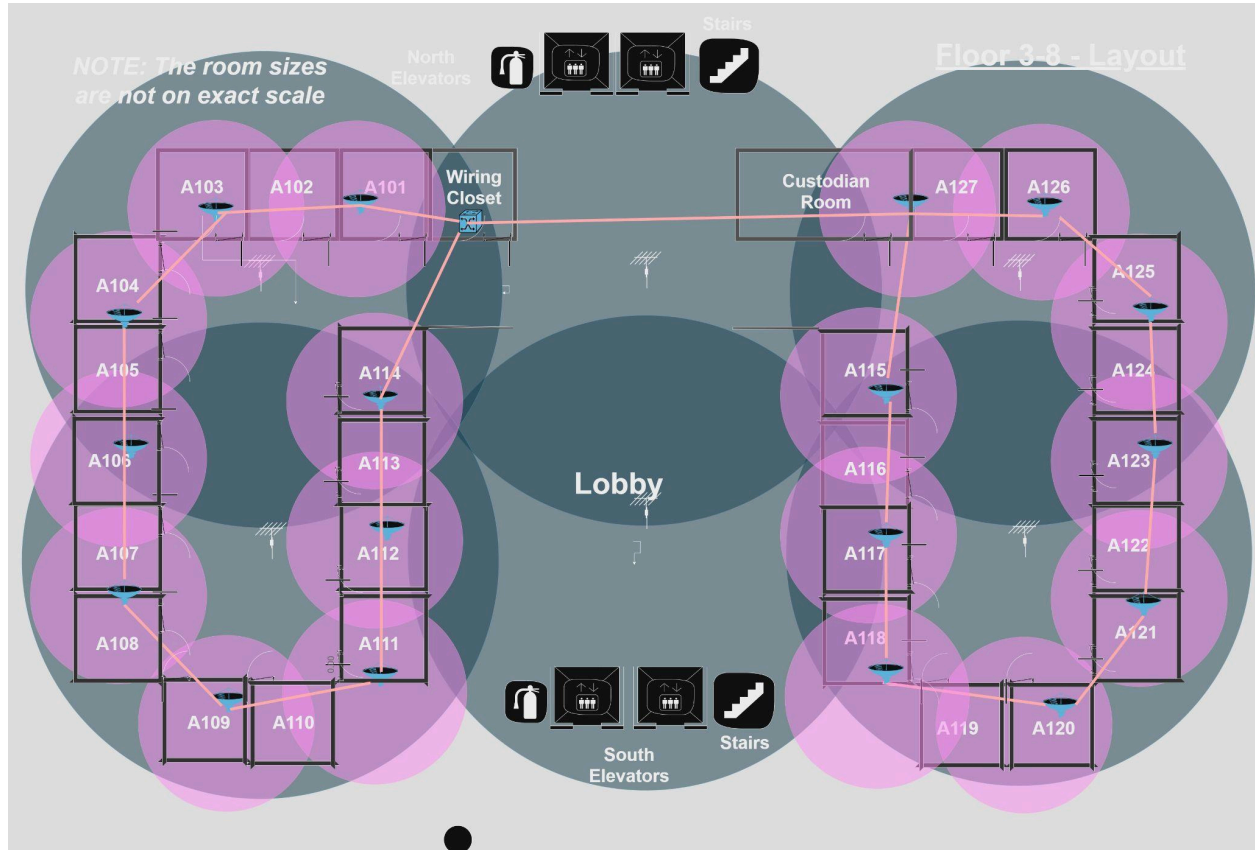


Layout of Floor 2

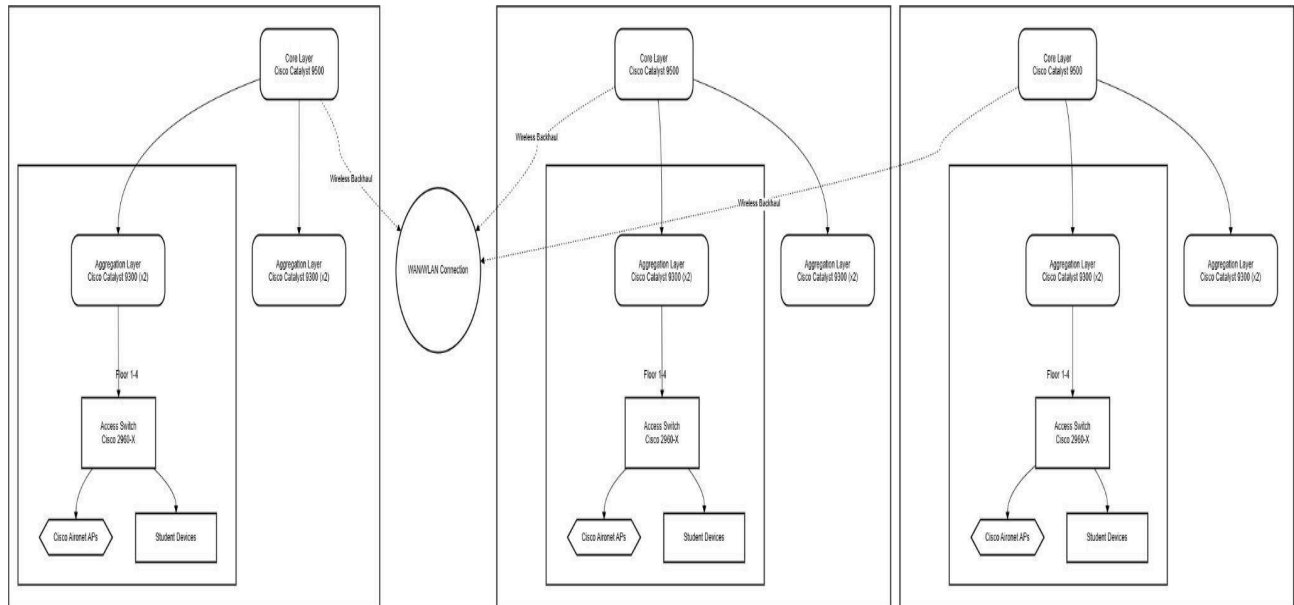




Layout of Floors 3 to 8



Network Architecture/Topology



Open as an image and explain.

-  **Core Switch (Main Brain):**
A powerful central switch (like a "traffic controller") that connects everything in the campus. Think of it as the school's main office.
-  **Aggregation Switches (Middle Managers):**
Two backup switches that distribute internet/network access to different parts of the campus. Like hallway monitors directing traffic to classrooms.
-  **Access Switches (Classroom Hubs):**
Smaller switches in each floor/area connecting wired devices (computers, printers) and wireless access points (APs).
-  **Wireless Access Points (WiFi Towers):**
Devices like "Cisco Aironet" that broadcast WiFi for students' phones, laptops, etc.
-  **Student Devices:**
Wired computers and wireless devices (laptops, phones) used in classrooms.



1. Core Components

- **Servers:**
 - *File Server*: Stores all shared files (e.g., student records, lesson plans).
 - *Email Server*: Handles campus emails.
 - *Web Server*: Hosts the school/campus website.
 - *DHCP/DNS Server*: Assigns IP addresses and directs internet traffic (like a phonebook for devices).
- **Storage Systems:**
 - *NAS (Network-Attached Storage)*: A giant shared hard drive for backups.
 - *SAN (Storage Area Network)*: High-speed storage for critical data (like exam databases).
- **Network Switches:**
 - High-powered switches (like **Cisco Catalyst 9500**) connect the servers to the rest of the campus network.

2. The Diagram Connections

In our campus network diagram:

- The server room connects directly to the **Core Layer Switch** (the "main brain" switch).
- All requests from devices (e.g., a student accessing a file or a teacher printing) pass through the server room.

The WAN acts as a "**digital highway**" that links geographically separated campuses, allowing them to:

- Share resources (files, servers, internet).
- Communicate securely (e.g., video calls between campuses).
- Centralize management (e.g., one IT team managing all campuses)

Each campus has a **Core Layer Switch** (e.g., Cisco Catalyst 9500). These switches connect to the WAN via:

- **WAN Routers**: Dedicated routers at each campus (not explicitly shown in the diagram) that link to the WAN cloud.



Example Flow:

Student Laptop → Access Switch → Aggregation Switch → Core Switch → Server Room → Back to Student

3. Redundancy & Safety

- **Backup Power:**
 - UPS (Uninterruptible Power Supply): Battery backup to keep servers running during power outages.
 - Generator: For longer outages.
- **Cooling Systems:**
 - AC units or fans to prevent overheating (servers get hot!).
- **Data Backup:**
 - Automatic backups to external drives or cloud storage (in case of fire/floods).

4. Security

- **Physical Security:**
 - Locked door, biometric access (fingerprint scanners), and CCTV cameras.
- **Digital Security:**
 - Firewall: Acts as a "security guard" blocking hackers.
 - Encryption: Scrambles data so only authorized users can read it.



Explanation

1. Hardware

Core Router: Cisco Catalyst 8200L-1N-4T Router (\$1,047.99)

We chose this router for its branch/campus edge connectivity with high throughput and modular interfaces, which supports SD-WAN, VPN tunnels, and secure inter-campus connections to UH Sugar Land and UH Clear Lake, connecting these 2 campuses to the main campus' core/network backbone.

Layer 3 Switch (48 port): Cisco Catalyst 1300-48P-4X (\$1,663.99)

This switch handles inter-VLAN routing, DHCP relay, and ACLs, with 48 PoE+ ports for powering APs and IP devices. Layer 3's functionality is crucial to segment the student, admin, facilities, and guest VLANs and is best for core switching and local routing in the building's MCF.

Access Switch (24-port): Cisco Catalyst 9200 (\$2,946.99)

The access switch is designed for layer access. It supports PoE+, 802.1X authentication, and QoS for video and voice traffic. The reason we are choosing this is because it is affordable and scalable for IDF closets on each floor

Wireless LAN Controller: Cisco WAN Network Interface Module (\$1,751.99)

This hardware provides centralized control of all APs, seamless roaming, and integrates with the core router. It will be used to manage Wi-Fi across all 8 floors with minimal config effort per AP.

Firewall: Cisco FirePOWER 1120 ASA (\$1,005.99)

This firewall is suitable for URL filtering and malware defense and medium enterprises like this. It also can enforce segmentation between the VLANs and block guest access, improving security between the data centers.



Wi-Fi Access Point: Cisco Business 140AC (\$89.99)

This is one of Cisco's smaller, more affordable access points, but it can still be used for common use cases, especially in dorms, offices, and mid-size networks. Additionally, it supports both 2.4 GHz and 5 GHz frequency bands (dual-band Wi-Fi), which works well for students studying or need to hop on a Zoom call.

Rack Server: Cisco UCS C220 M7 (\$3,456.99)

This high-performance server has 2 socket racks for running active directory, DHCP, DNS, and backup agents. Virtual machines are hosted for student and staff services, with scalable RAM and storage, along with Cisco UCS integration. This device is best for local data centers and local services authentication.

Storage Server: Cisco UCS S3260 (\$13,158.99)

We are using a high-capacity storage node for these main components: backups, file sharing and logs, and future-proof storage growth. This is optimized enough for data durability, redundant power, and modular drives. It is also scalable and is a reliable onsite storage solution for each building.

Fiber Optic Cable: Cisco 100GBase Direct Attach Cable (7m) (\$551.99)

This cable supports 100 Gbps speed for core uplinks between MDF, IDF, and servers. With short distance direct attach, it offers high-speed connectivity in the rack and is very cost-effective. We can consider it an ideal choice for connections between the core switch, router, firewall, and servers. It is also best suited for core to core links in the data center and uplinks within the building.



2. Subnet

We divided the network into these subsections:

Subnet	CIDR	Subnet Mask	Gateway IP	IP Range	Broadcast Address
Admin Office Staff	172.16.0.0/24	255.255.255.0	172.16.0.1	172.16.0.1 - 172.16.0.254	172.16.0.255
Student Residents	172.16.1.0/21	255.255.248.0	172.16.1.1	172.16.1.1 - 172.16.8.254	172.16.2.255
Facilities / Maintenance	172.16.3.0/24	255.255.255.0	172.16.3.1	172.16.3.1 - 172.16.3.254	172.16.3.255
Visitor Wi-Fi	192.168.0.0	255.255.255.0	192.168.0.1	192.168.0.1 - 192.168.0.254	192.168.0.255

Table 1. Subnets breakdown with CIDR, subnet mask, gateway IP, IP range, broadcast address for each subnet section.

Based on the RFC 1918 private IP block, the Administration Office Staff, Student Residents, and Facilities/Maintenance will be using the 172.16.0.0 network while the Visitor Wi-Fi is using the 192.168.0.0. Besides the Visitor Wi-Fi, which needs to use a separate network for guest traffic control and security, the remaining three subnets use the same private network. 172.16.0.0 allows for a unified internal address space.

The Admin Office Staff uses the /24 subnet mask as it is the standard size subnet. With 256 total IP addresses, breaking the number down, we have 1 network address, 254 usable addresses, and 1 broadcast address. By using this, we can avoid wasting IPs and provide flexibility while still leaving room for future expansion. These are also enough for a small number of admin staff computers.

For the Student Residents subnet, we have to account for the total number of devices per student that need to access the network. Suppose each bedroom is shared by 2 students; the total students in 27 two-bedroom apartments across Floor 3-8 is going to be 648 students. Based on this number, the subnet mask we are using is the /21, assuming that each student has 3 devices that need to be connected. We will need at least 1944 usable IPs, and the /21 subnet can store up to 2048 IPs. This is a safe amount of IPs we need to account for, although there will be a lot of



unused IPs. However, the /22 subnet is not enough to safely account for 1944 accessed IPs, knowing that it can only store 1022 usable IP addresses.

We only need to account for staff usage in the Facilities and Maintenance subnet, which is the main reason why we are choosing the /24. This includes 254 usable IP addresses, enough for devices like smart building systems, HVAC, cameras, and badge readers. Using a standard size subnet for these IT devices can reduce subnet complexity.

As explained at the beginning of this section, the Visitor Wi-Fi will be using a different network from the rest of the building. We are choosing the 192 network because it is small and isolated enough for untrusted devices. Besides that, this is also a commonly used network for coffee shops or visitor zones. Using this can also imply easy firewall rules and separation.

The gateway IP and IP ranges for every subnet remain the same. However, for the student residents' subnet, the IP will have to range from 172.16.1.1 to 172.16.8.254 to account for 2046 usable IP addresses. We assigned the .1 end for the gateway IP; it is a common convention and is very simple and easy for the admin to remember. The IPs will range from the gateway IP, .1 to .254, based on the usable IP addresses. The .255 end is the broadcast address. It is for all of the devices on the subnet to communicate with each other and is unusable.

3. How IP and subnet design support cross-campus communication

The address spaces given are 172.16.0.0/18 for internal staff, residents, and facility networks; and 192.168.0.0/24 for guest Wi-Fi. With a large private block (/18), we have 16,384 total IP addresses that can be divided consistently across all three campuses. An example for this allocation is 172.16.0.0/20 per campus. With this, each campus gets 4,096 IPs, and subnet divisions are used identically on each site. This is to ensure that there are no overlapping IPs, predictable routing, and easier troubleshooting and access control. The IP design is also suited for routing because IP blocks are not overlapping. We can also use dynamic routing to advertise subnets between campuses.



Cost Assessment

1. Hardware List & Pricing

	Item	Model	Qty	Unit Price	Total
Network Hardware	Core Router	Cisco Catalyst 8200L 1N 4T	1	\$1,047.99	\$1,047.99
	Layer 3 Switch (48-port)	Cisco Catalyst 1300 48P 4X	10	\$1,663.99	\$16,639.90
	Access Switch (24-port)	Cisco Catalyst 9200	16	\$2,946.99	\$47,151.84
	Wireless LAN Controller	Cisco WAN Network Interface Module	1	1,751.99	\$1,751.99
	Firewall	Cisco FirePOWER 1120 ASA	3	1,005.99	\$3,017.97
APs	Wi-Fi Access Point	Cisco Business 140AC	516	\$89.99	\$46,434.84
Servers	Rack Server	Cisco UCS C220 M7	2	\$3,456.99	\$6,913.98
	Storage Server	Cisco UCS S3260	1	\$13,158.99	\$13,158.99
Cabling	Fiber Optic Cable	Cisco 100GBase Direct Attach Cable (7m)	150	\$551.99	\$82,798.50
TOTAL					\$218,916.00

Table 2. Hardware, APs, servers, and cabling list with item, model, quantity, unit price, and total pricing.

2. Total Budget & Cost Breakdown

The devices used here are all from the Cisco brand and of high quality, which is what contributed to the overall cost being under the total budget provided of \$500,000 for the overall network infrastructure and design of CougarTech Living Hall. The devices that were used include the cables that were laid down throughout the offices on Floor 1 and along the outside walls of all the rooms from Floors 2-8. This cable made up the majority of the cost due to how much was needed to satisfy the amount of area that had to be covered for the dimensions of the building. For the rest of the devices, a substantial portion are housed in the MDF, with the exceptions being one switch on Floors 2-8, as well as the number of access points on each of the floors.



Safety / Security

To enhance security, performance, and network segmentation, VLANs are used to separate traffic by user type and service. Each VLAN is assigned a unique IP subnet and purpose. Admin (10), Residents (20), Visitors (30), Voice (40), Backup (50), and Servers (60) allowing controlled communication, better traffic management, and easier troubleshooting.

Traffic between campuses is encrypted (e.g., IPsec VPN) to prevent eavesdropping. Each campus's core switch/router filters traffic entering/exiting the WAN.

Backup & Recovery Strategy

CougarTech Hall will be using a layered backup and recovery approach to ensure data safety and ransomware protection. A Synology NAS with RAID 5 stores daily incremental and weekly full backups of critical systems, including Active Directory, DHCP, and server configurations. A secondary external USB drive provides offline air-gapped backups, while encrypted cloud storage offers offsite protection.

To defend against ransomware, the system uses immutable snapshots, encrypted backups, role-based access control, and antivirus scanning on the NAS. A Smart UPS ensures safe shutdown during power loss, and alerts are configured for backup failures or suspicious activity.



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